



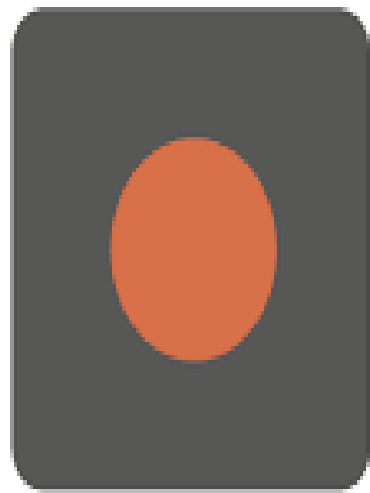
BITS Pilani

Introduction to Statistical Methods





Session 10
Analysis of Variance (ANOVA)
(05/04/2026)



Start Recording

Work Integrated Learning Programmes



Analysis of Variances (ANOVA)

<http://faculty.washington.edu/heagerty/Books/Biostatistics/TABLES/F-Tables/>

Study the following example

Example 1:

A company has developed three drying formulae for curing glue.

How to decide which formulae is better in curing the glue?

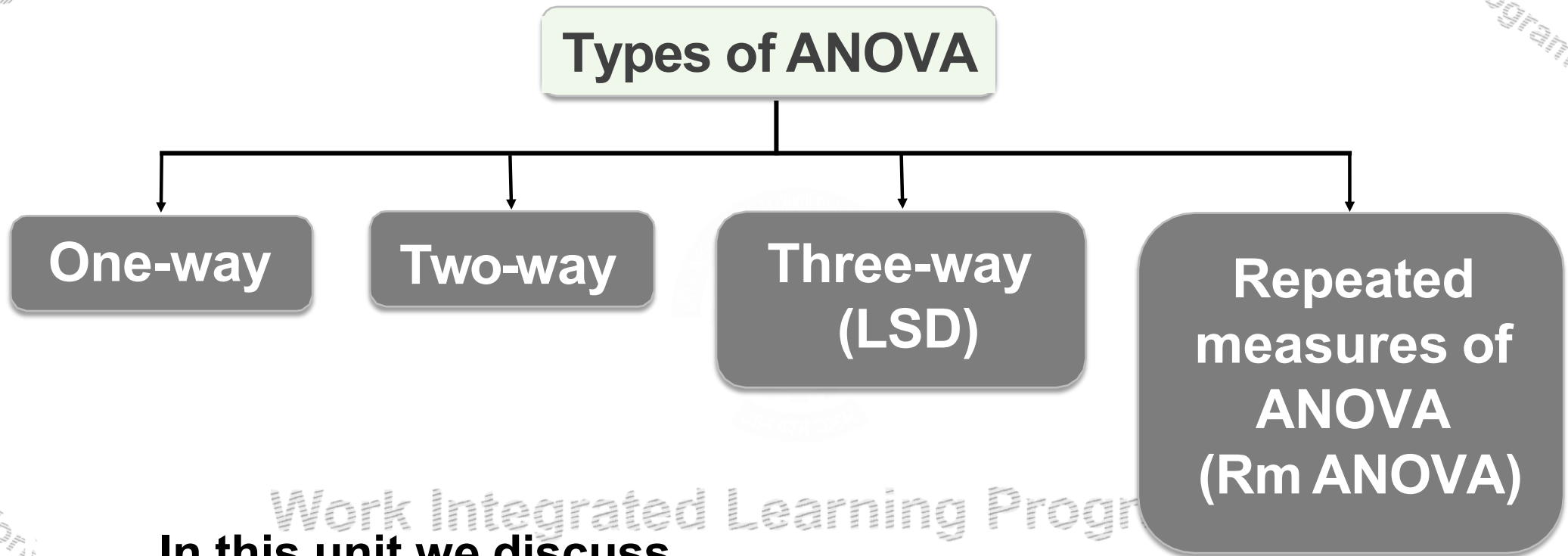
Suppose some data is given below, now how will you decide which formulae may be better?

Formula A	13	10	8	11	8	
Formula B	13	11	14	14		
Formula C	4	1	3	4	2	4

Why Analysis of Variance (ANOVA)?

- We could compare the means, one by one using t-tests for difference of means.
- **Problem:** each test contains type I error
- The total type I error is $1-(1-\alpha)^k$ where k is the number of means.
- For example, if there are 5 means and $\alpha=0.05$, we must make 10 two by two comparisons.
- Thus, the type I error is $1-(0.95)^{10}$, which is 0.4012.
- That is, 40% of the time you will reject the null hypothesis of equal means in favor of the alternative!
- ANOVA used to test equality of more than two population means against not equal.

Testing of Hypothesis → Analysis of Variance (ANOVA)



In this unit we discuss only One-way ANOVA and Two-way ANOVA in detail

Analysis of Variance

- Analysis of Variance (ANOVA) can be used to test for the equality of three or more population means
- We want to use the sample results to test the following hypotheses:
$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$
$$H_1: \text{Not all population means are equal}$$
- If H_0 is rejected, we cannot conclude that all population means are not equal
- Rejecting H_0 means that at least two population means have different values

Analysis of Variance

Assumptions for Analysis of Variance

- For each population, the response (dependent) variable is normally distributed
- The variance of the response variable, denoted σ^2 , is the same for all of the populations
- The observations must be independent



One-way Analysis of Variance

In one-way ANOVA the effect of one factor on the mean is tested. It is based on independent random samples drawn from k – different levels of a factor, also called **treatments(groups)**.

The data and the notations used in one-way ANOVA are represented in the following tabular structure.

One-way Analysis of Variance



Treatment-1	Treatment-2	Treatment-3	.	.	Treatment-k
x11	x21	x31	.	.	xk1
x12	x22	x32	.	.	xk2
x13	x23	x33	.	.	xk3
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
x1n ₁	x1n ₂	x1n ₃	.	.	x1n _k
C ₁	C ₂	C ₃	.	.	C _k

$$n_1 + n_2 + n_3 + \dots + n_k = n$$

$$C_1 + C_2 + C_3 + \dots + C_k = G$$

**The hypotheses to
be tested are**

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

VS

$$H_1 : \mu_1 \neq \mu_2 \neq \dots \neq \mu_k$$

One-way Analysis of Variance

- Assume that a simple random sample of size n_j has been selected from each of the k populations or treatments. For the resulting sample data, let

x_{ij} = value of observation i for treatment j

n_j = number of observations for treatment j

$\bar{x}_j$ = sample mean for treatment j

s_j = sample standard deviation for treatment j

Between-Treatments Estimate of Population Variance σ^2

- The estimate of σ^2 based on the variation of the sample means is called the mean square due to treatments and is denoted by MSTR

$$\text{MSTR} = \frac{\sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2}{k - 1}$$

Denominator is the degrees of freedom associated with SSTR

Numerator is called the sum of squares due to treatments (SSTR)

Within-Treatments Estimate of Population Variance σ^2

- The estimate of σ^2 based on the variation of the sample observations within each sample is called the mean square error and is denoted by MSE

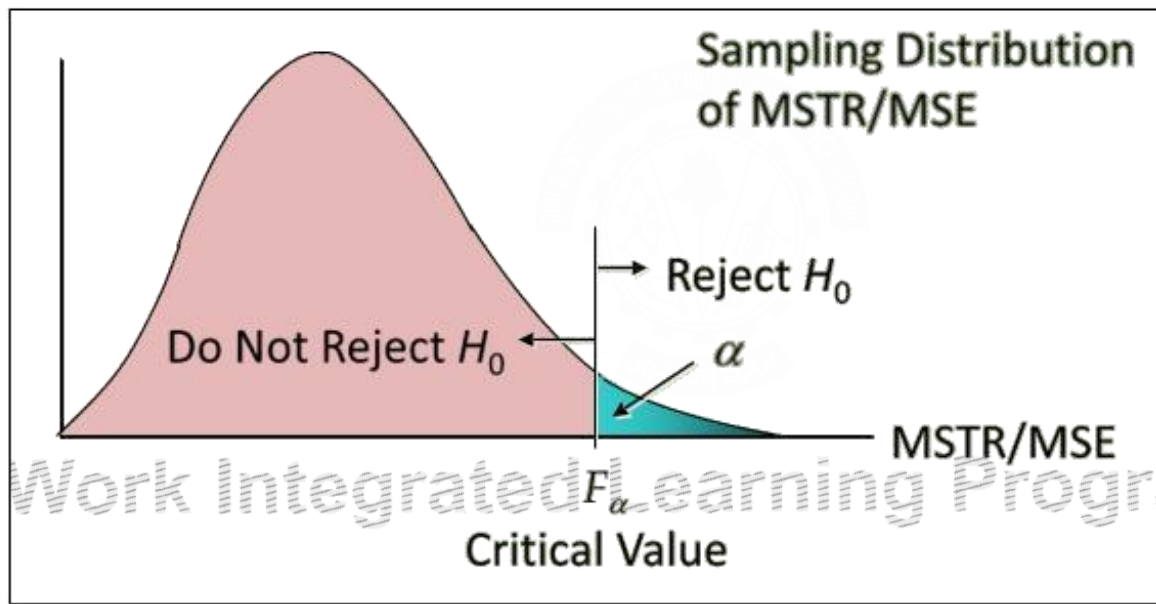
$$\text{MSE} = \frac{\sum_{j=1}^k (n_j - 1)s_j^2}{n - k}$$

Denominator is the degrees of freedom associated with SSE

Numerator is called the sum of squares due to error (SSE)

Fisher's F-test

- If the null hypothesis is true and the ANOVA assumptions are valid, the sampling distribution of $MSTR/MSE$ is an F distribution with $MSTR$ d.f. equal to $k - 1$ and MSE d.f. equal to $n - k$.



ANOVA table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SSTR	$k - 1$	$MSTR = \frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$
Error	SSE	$n - k$	$MSE = \frac{SSE}{n-k}$	
Total	SST	$n - 1$		

SST is partitioned into SSTR and SSE.

SST's degrees of freedom (d.f.) are partitioned into SSTR's d.f. and SSE's d.f.

- SST divided by its degrees of freedom $n - 1$ is the overall sample variance that would be obtained if we treated the entire set of observations as one data set.
- With the entire data set as one sample, the formula for computing the total sum of squares, SST, is:

$$SST = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x})^2 = SSTR + SSE$$

One-way ANOVA example



Test at 5% level of significance is there any significant difference in mean of the three samples obtained from the Normal population of equal variances

Group 1	Group 2	Group 3
8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_1 : \mu_1 \neq \mu_2 \neq \mu_3$$

For $\alpha=0.05$ F tab value for $(k-1, n-k) = (2, 12)$ d.o.f. is 3.89

Solution Method-1:

	Group 1	Group 2	Group 3
Grand total, G = 150	8 10 7 14 11	7 5 10 9 9	12 9 13 12 14
Total C _i	50 $\bar{x}_1 = 10$	40 $\bar{x}_2 = 8$	60 $\bar{x}_3 = 12$

Step 1: $H_0: \mu_1 = \mu_2 = \mu_3$, $H_1: \mu_1 \neq \mu_2 \neq \mu_3$

Step 2: **SSTR**: Sum of sq. of variations between treatments
 $= r \sum (\bar{x}_j - \bar{\bar{x}})^2 = 5[(10-10)^2 + (8-10)^2 + (12-10)^2] = 5(0+4+4) = 40$

Step 3: **SSE**: Sum of sq. of variations due to error $= \sum \sum (x_{ij} - \bar{x}_j)^2$
 $= (8-10)^2 + (10-10)^2 + (7-10)^2 + (14-10)^2 + (11-10)^2 +$
 $(7-8)^2 + (5-8)^2 + (10-8)^2 + (9-8)^2 + (9-8)^2 +$
 $(12-12)^2 + (9-12)^2 + (13-12)^2 + (12-12)^2 + (14-12)^2 = 60$

Step 4: **SST** Total variation = SSTR + SSE = 100

Step 5: ANOVA Table

Grand total , G = 150 and $\bar{\bar{x}} = 10$

One-way ANOVA table

Source of variation	df	Sum of squares	Mean sum of squares	F-ratio
Between groups	$k - 1 = 2$	SSTR =40	$MSTR = \frac{SSTR}{k - 1} = 20$	$F = \frac{SSTR}{SSE} = 4$
Within groups	$n - k = 12$	SSE =60	$MSE = \frac{SSE}{n - k} = 5$	
Total		SST =100	$F \approx F$ - distribution with $k - 1$ and $n - k$ df	

For $\alpha = 0.05$ F tab value for $(k-1, n-k) = (2, 12)$ d.o.f. is 3.89. Thus, $F_{\text{tab}} 3.89 < F_{\text{cal}} 4$, Reject H_0 and accept H_1 .

Solution Method-2: using Correction Factor

	Group 1	Group 2	Group 3
Grand total, G = 150	8	7	12
	10	5	9
	7	10	13
	14	9	12
	11	9	14
Total C _i	50	40	60

Grand total , G =150

Step 1: Correction Factor (CF)= $G^2 / n = (150)^2 / 15 = 1500$

Step 2: SST = $\sum \sum x_{ij}^2 - CF = 1600 - 1500 = 100$

Step 3: SSTR = $\frac{\sum C_i^2}{n_i} - CF = \frac{(50)^2}{5} + \frac{(40)^2}{5} + \frac{(60)^2}{5} - 1500 = 40$

Step 4: SSE = SST - SSTR = $100 - 40 = 60$

Step 5: ANOVA Table



One-way ANOVA table

Source of variation	df	Sum of square	Mean sum of squares	F-ratio
Between groups	$k - 1 = 2$	$SSTR = 40$	$MSTR = \frac{SSTR}{k - 1} = 20$	$F = \frac{SSTR}{SSE} = 4$
Within groups	$n - k = 12$	$SSE = 60$	$MSE = \frac{SSE}{n - k} = 5$	
Total		$SST = 100$	$F \approx F\text{-distribution with } k - 1 \text{ and } n - k \text{ df}$	

For $\alpha=0.05$ F tab value for $(k-1, n-k) = (2, 12)$ d.o.f. is 3.89. Thus $F_{\text{tab}} 3.89 < F_{\text{cal}} 4$, Reject H_0 and accept H_1 .

One-way ANOVA Example-2:



Test at 5% level of significance is there any significant difference in mean iron intake among four groups of patients? Also test, if significant, which group means have contributed to the difference in means using LSD test

A clinical trial Iron intake of four groups of patients (mg)

Group 1	Group 2	Group 3	Group 4
11.5	19.5	18.5	30.0
12.5	18.5	16.5	26.5
18.5	16.0	24.5	27.0
21.0	22.0	30.0	34.0
28.0	30.0	28.5	20.0
26.0	24.5	14.0	22.5
14.0	19.0	19.0	28.0
22.0	24.0	17.0	32.0
20.0	19.5	18.0	27.0
22.0	15.0	29.0	25.5

Solution:

Grand total , $G = 981$

Step 1: Correction Factor (CF) = $G^2 / n = (981)^2 / 40 = 19847.03$

Step 2: SST = $\sum \sum x_{ij}^2 - CF = 21119.5 - 19847.03 = 1272.47$

Step 3: SSTR = $\frac{\sum C_i^2}{n_i} - CF = 20196.6 - 19847.03 = 349.52$

Step 4: SSE = SST – SSTR = 1272.47 - 349.52 = 922.95

Step 5: ANOVA Table

One-way ANOVA table



Source of variation	df	Sum of squares	Mean sum of squares	F-ratio
Between groups	3	349.52	116.51	F=4.455
Within groups	36	922.95	25.64	
Total	39	1272.47	F(4.46) > F _(0.05; 3,36) = 4.38	

H_0 rejected and H_1 accepted

One-way ANOVA Example-3:



Three drying formulas for curing glue are studied



Formula A	13	10	8	11	8	
Formula B	13	11	14	14		
Formula C	17	14	13	10	11	12

Test at 5% level of significance whether is any difference in the mean curing time of glue?

Solution:

Grand total , $G = 179$

Step 1: Correction Factor (CF) = $G^2 / n = (179)^2 / 15 = 2136.067$

Step 2: SST = $\sum \sum x_{ij}^2 - CF = 2219.5 - 2136.067 = 82.93$

Step 3: SSTR = $\frac{\sum C_i^2}{n_i} - CF = 2164.167 - 2136.067 = 28.1$

Step 4: SSE = SST – SSTR = $82.93 - 28.1 = 54.83$

Step 5: ANOVA Table

One-way ANOVA table



Source of variation	df	Sum of squares	Mean sum of squares	F-ratio
Between groups	2	28.1	14.05	F=3.078
Within groups	12	54.83	4.569	
Total	14	82.93	F(3.078) < F _(0.05; 2, 12) = 3.885	

H₀ not rejected and **H₁ rejected**. **P – value = 0.084**



Two-way Analysis of Variance (Two-way ANOVA)

Work Integrated Learning Programmes



TWO-WAY ANOVA

- In two-way ANOVA, a study variable is compared over three or more groups, controlling for another variable.
- The grouping is taken as one factor and the control is taken as another factor. The grouping factor is usually known as **Treatment**.
- The control factor is usually called **Block**.
- The accuracy of the test in two-way ANOVA is considerably higher than that of one-way ANOVA, as the additional factor, block is used to reduce the error variance.

Treatments

Blocks

	1	2	...	k	Total
1	x_{11}	x_{12}	...	x_{1k}	$x_{.1}$
2	x_{21}	x_{22}	...	x_{2k}	$x_{.2}$
3	x_{31}	x_{32}	...	x_{3k}	$x_{.3}$
...	...	...	...	...	...
m	x_{m1}	x_{m2}	...	x_{mk}	$x_{.m}$
Total	$x_{1.}$	$x_{2.}$	...	$x_{k.}$	G

Steps involved in two-way ANOVA



Step 1: In two-way ANOVA we have two pairs of hypotheses, one for treatments and one for the blocks.

Null Hypotheses

- H_01 : There is no significant difference among the population means of different Treatments(groups)
- H_02 : There is no significant difference among the population means of different Blocks

Alternative Hypotheses

- H_11 : At least one pair of treatment means differs significantly
- H_12 : At least one pair of block means differs significantly

Steps involved in two-way ANOVA



Step 2: Data is presented in a rectangular table form as described in one-way ANOVA.

Step 3: Level of significance α

Step 4: Test Statistic

For treatments

- $$F = \frac{MSTR}{MSE}$$

For blocks

- $$F = \frac{MSBL}{MSE}$$

Steps involved in two-way ANOVA



To find the test statistic we have to find the following intermediate values.

- i) Correction Factor: $C.F. = \frac{G^2}{n}$, where, $G = \sum_{j=1}^m \sum_{i=1}^k x_{ij}$
- ii) Total Sum of Squares: $SST = \sum_{i=1}^k \sum_{j=1}^m x_{ij}^2 - C.F.$
- iii) Sum of Squares between Treatments: $SSTR = \sum_{i=1}^k \frac{x_i^2}{m} - C.F.$
- iv) Sum of Squares between blocks: $SSBL = \sum_{j=1}^m \frac{x_j^2}{k} - C.F.$
- v) Sum of Squares due to errors: $SSE = SST - SSTR - SSBL$

Steps involved in two-way ANOVA



Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SSTR	k-1	MSTR= SSTR/k-1	F= MSTR/MSE And F= MSBL/MSE
Blocks	SSBL	(m-1)	MSBL = SSBL/m- 1	
Error	SSE	(k-1)(m-1)	MSE= SSE/(k-1)(m-1)	
Total	SST	n-1		

Two-way ANOVA :

Example-1



A Farmer applies three types of fertilizers on four separate plots. The figures on yield per acre are tabulated. Test at 5% level of significance is there any significant difference in the yield due to fertilizers and also due to the plots

Plots→ Fertilizers	A	B	C	D
Nitrogen	6	4	8	6
Potash	7	6	6	9
Phosphate	8	5	10	9

H_0 : Plots do not differ the yield and Fertilizers do not differ the yield

H_1 : Plots differ the yield and Fertilizers differ the yield

Solution:

Plots(Treatments) Fertilizers(Blocks)	A	B	C	D	Total
Nitrogen	6	4	8	6	24
Potash	7	6	6	9	28
Phosphate	8	5	10	9	32
Total	21	15	24	24	G=84

Step 1: $CF = G^2 / n = (84)^2 / 12 = 588$

Step 5: $SSE = SST - SSTR - SSBL = 36 - 18 - 8 = 10$

Step 2: $SST = \sum \sum x_{ij}^2 - CF = 624 - 588 = 36$

Step 6: ANOVA Table

Step 3: For column SSTR $= \frac{\sum C_i^2}{n_i} - CF = \frac{(21)^2}{3} + \frac{(15)^2}{3} + \frac{(24)^2}{3} + \frac{(24)^2}{3} - 588 = 18$

Step 4: For row SSBL $= \frac{\sum R_i^2}{n_i} - CF = \frac{(24)^2}{4} + \frac{(28)^2}{4} + \frac{(32)^2}{4} - 588 = 8$

Two-way ANOVA Table



Source of variation	df	Sum of squares	Mean sum of squares	F-ratio
Between Treatments (Plots)	$k - 1 = 3$	$SSTR = 18$	$MSTR = SSTR / k - 1 = 6$	For Treatment $F = MSTR / MSE = 6 / 1.67 = 3.6$ For Block $F = MSBL / MSE = 4 / 1.67 = 2.4$
Between Blocks (Fertilizers)	$m - 1 = 2$	$SSBL = 8$	$MSBL = SSBL / m - 1 = 4$	
Within groups	$(k - 1) * (m - 1) = 6$	$SSE = 10$	$MSE = SSE / (k - 1) * (m - 1) = 1.67$	

F tab value for treatment = (3, 6) d.o.f. is 4.76 and thus $F_{cal} < F_{tab}$ hence Accept H_0

F tab value for block = (2, 6) d.o.f. is 5.14 and thus $F_{cal} < F_{tab}$ hence Accept H_0

Maximum Likelihood Estimation (MLE)

- The method of maximum likelihood was first introduced by R. A. Fisher, a geneticist and statistician, in the 1920s.
- Most statisticians recommend this method, at least when the sample size is large, since the resulting estimators have certain desirable efficiency properties.
- Maximum likelihood estimation(MLE) is a method to find most likely density function, that would have generated data.
- MLE requires one to make distribution assumption first.

Maximum Likelihood Estimation (MLE)



- Estimation is the process of estimating unknown true values of population parameters using their corresponding best sample statistics (good estimators) in an optimum manner.
- An estimator is said to be a good if it is
 - unbiased,
 - consistent,
 - efficient and
 - sufficient while estimating its parameter.

Maximum Likelihood Estimation (MLE)



- Method of Maximum Likelihood Estimation is the best and most popular one among all methods to obtain an almost good or best estimator for a population parameter.
- It is a method of obtaining an estimator which most (maximum) likely estimates the true value of the parameter i.e., finding an estimator that can give most likely nearer value for the unknown true value of parameter.
- The corresponding estimator is called **Maximum Likelihood Estimator (MLE)**.

MLE: Example



- A sample of ten new bike helmets manufactured by a certain company is obtained. Upon testing, it is found that the **first, third, and tenth helmets are flawed**, whereas the others are not.

- Let $p = P(\text{flawed helmet})$
- Define (Bernoulli) random variables $X_1, X_2, \dots, X_{10}$ by

$$X_1 = \begin{cases} 1 & \text{if 1st helmet is flawed} \\ 0 & \text{if 1st helmet isn't flawed} \end{cases} \quad \dots \quad X_{10} = \begin{cases} 1 & \text{if 10th helmet is flawed} \\ 0 & \text{if 10th helmet isn't flawed} \end{cases}$$

Source: Probability and Statistics for Engineering and the Sciences, Jay L Devore, 8th Ed, Cengage

MLE: Example



- Then for the obtained sample, $X_1 = X_3 = X_{10} = 1$ and the other seven X 's are all zero.
- The probability mass function (pmf) of any particular X , is $p^x \cdot (1 - p)^{(1-x)}$, which becomes p if $x = 1$ and $1-p$ when $x = 0$.

As the conditions of various helmets are independent of one another, this implies that the X 's are independent, so their joint probability mass function is the product of the individual pmf's.

Joint pmf evaluated at the observed X 's (likelihood function) is

$$f(x_1, \dots, x_{10}; p) = p(1 - p)p \dots p = p^3(1 - p)^7$$

MLE: Example



$$f(x_1, \dots, x_{10}; p) = p(1 - p)p \dots p = p^3(1 - p)^7$$

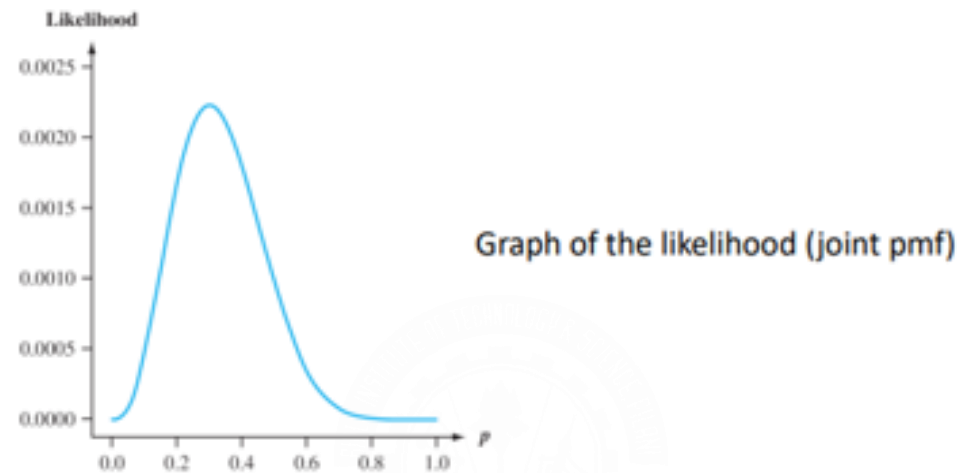
Suppose that $p = .25$. Then $f = (.25)^3(.75)^7 = .002086$.

If instead $p = .50$, then $f = (.50)^3(.50)^7 = .000977$.

For what value of p is the obtained sample most likely to have occurred?

That is, for what value of p is the joint probability mass function (likelihood function) as large as it can be?

MLE: Example



- Figure shows a graph of the likelihood as a function of p .
- It appears that the graph reaches its peak above $p = 0.3$
= the proportion of flawed helmets in the sample.

MLE: Example



We can verify our visual impression by using calculus to find the value of p that maximizes likelihood.

Working with the natural log of the joint pmf is often easier than working with the joint pmf itself, since the joint pmf is typically a product so its logarithm will be a sum.

Here, $\ln f = \ln[p^3(1 - p)^7]$
gives $\ln f = 3\ln(p) + 7\ln(1 - p)$

Differentiating and equating this derivative to 0 and solving for p gives $p = 3/10 = 0.30$ as conjectured.

That is, our point estimate is $p = .30$.

It is called the **maximum likelihood estimate** because it is the parameter value that maximizes the likelihood (joint pmf) of the observed sample.

MLE: Example



Suppose that rather than being told the condition of every helmet, we had only been informed that three of the ten were flawed.

Then we would have the observed value of a binomial random variable X = the number of flawed helmets.

– The pmf of X becomes $^{10}C_3 p^3(1 - p)^7$. The binomial coefficient ($^{10}C_3$) is irrelevant to the maximization, so again $p = 0.30$.

The likelihood function tells us how likely the observed sample is as a function of the possible parameter values.

Maximizing the likelihood gives the parameter values for which the observed sample is most likely to have been generated—that is, the parameter values that "agree most closely" with the observed data.



Maximum Likelihood Estimation (MLE)

Suppose we have a random sample $x_1, x_2, \dots, x_n$ whose assumed probability distribution depends on some unknown parameter θ

Ex:

- 1) For Bernoulli unknown parameter are p .
- 2) For Binomial unknown parameters are n, p .
- 3) For Poisson unknown parameter is λ
- 4) For Normal unknown parameters are μ and σ^2 .

Our goal is to find good estimate of θ (population parameter) using sample and which can be done with the help of MLE.

Maximum Likelihood function



It is observed that a good estimate of unknown parameter θ would be the value of θ that maximizes the probability.

i.e. the likelihood of getting the data we observed (this is reason, why we called as likelihood function).

Let $x_1, x_2, \dots, x_n$ be i.i.d. random variables drawn from some probability distribution that depends on some unknown parameter θ .

$$L(\theta) = \prod_{i=1}^n f(x_i / \theta)$$

The goal of MLE is to maximize likelihood function

$$\begin{aligned} L(\theta) &= f(x_1, x_2, \dots, x_n | \theta) \\ &= f(x_1 | \theta) * f(x_2 | \theta) * \dots * f(x_n | \theta) \end{aligned}$$

Maximum Likelihood Estimation (MLE)

- The maximum likelihood estimate (MLE) of θ is that value of θ which maximizes likelihood(θ).

- It is defined as

$$L(\theta) = \prod_{i=1}^n f(x_i / \theta)$$

$$\log L(\theta) = \sum_{i=1}^n \log f(x_i / \theta)$$

For maximization we have,

$$\frac{dL}{d\theta} = 0; \quad \frac{d^2L}{d\theta^2} < 0$$

Maximum Likelihood Estimation (MLE)



If L and $\log_e L$ are not differentiable or integrable or principle of maxima-minima fails then in such case direct method of finding the estimator of the parameter which maximizes L or $\log_e L$ is applied using order statistic principle empirically.

Work Integrated Learning Programmes



MLE for a Binomial distribution parameter

- Suppose we wish to find the MLE of θ for a Binomial distribution with k successes in n trials then,

$$p_k(k, \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

$$\log p_k(k, \theta) = \log \binom{n}{k} + k \log \theta + (n - k) \log(1 - \theta)$$

For extrema, substitute $\frac{\partial \log p_k(k, \theta)}{\partial \theta} = 0$, we get,

$$0 + \frac{k}{\theta} - \frac{n - k}{1 - \theta} = 0$$

$$k - k\theta = n\theta - k\theta \Rightarrow \theta = \frac{k}{n}$$

MLE for Poisson Distribution Parameter

- Suppose we have data generated from a Poisson distribution. We want to estimate the parameter of the distribution
- The probability of observing a particular random variable is $P(X; \mu) = \frac{e^{-\mu} \mu^X}{X!}$
- Joint likelihood by multiplying the individual probabilities together

$$P(X_1, X_2, \dots, X_n; \mu) = \frac{e^{-\mu} \mu^{X_1}}{X_1!} \times \frac{e^{-\mu} \mu^{X_2}}{X_2!} \times \dots \times \frac{e^{-\mu} \mu^{X_n}}{X_n!}$$

$$L(\mu; \mathbf{X}) = \prod_i e^{-\mu} \mu^{X_i}$$

$$L(\mu; \mathbf{X}) = e^{-n\mu} \mu^{n\bar{X}}$$

MLE for Poisson Distribution Parameter

- Note in the likelihood function the factorials have disappeared.
- This is because they provide a constant that does not influence the relative likelihood of different values of the parameter
- It is usual to work with the **log likelihood** rather than the likelihood.
- Note that maximising the log likelihood is equivalent to maximising the likelihood.

$$L(\mu; \mathbf{X}) = e^{-n\mu} \mu^{n\bar{X}}$$

$$\ell(\mu; \mathbf{X}) = -n\mu + n\bar{X} \log \mu$$

$$\frac{d\ell}{d\mu} = -n + \frac{n\bar{X}}{\mu}$$

$$\hat{\mu} = \bar{X}$$

Take the natural log of the likelihood function

Find where the derivative of the log likelihood is zero

Note that here the MLE is the same as the moment estimator

MLE for Exponential Distribution Parameter

- Suppose $X_1, X_2, \dots, X_n$ is a random sample from an exponential distribution with parameter λ . Because of independence, the likelihood function is a product of the individual pdf's:

$$\begin{aligned} f(x_1, \dots, x_n; \lambda) &= (\lambda e^{-\lambda x_1}) \cdot \dots \cdot (\lambda e^{-\lambda x_n}) \\ &= \lambda^n e^{-\lambda \sum x_i} \end{aligned}$$

- The natural logarithm of the likelihood function is

$$\ln[f(x_1, \dots, x_n; \lambda)] = n \ln(\lambda) - \lambda \sum x_i$$

MLE for Exponential Distribution Parameter

- Equating $(d/d\lambda)[\ln(\text{likelihood})]$ to zero results in $n/\lambda - \sum x_i = 0$, or $\lambda = n/\sum x_i = 1/\bar{x}$.
- Thus the MLE is $\hat{\lambda} = 1/\bar{X}$;

Work Integrated Learning Programmes

MLEs for Normal Distribution Parameters

- Let $X_1, \dots, X_n$ be a random sample from a normal distribution.
- The likelihood function is

$$f(x_1, \dots, x_n; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_1 - \mu)^2 / (2\sigma^2)} \cdot \dots \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_n - \mu)^2 / (2\sigma^2)}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\sum (x_i - \mu)^2 / (2\sigma^2)}$$

- so

$$\ln[f(x_1, \dots, x_n; \mu, \sigma^2)] = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

MLEs for Normal Distribution Parameters

- To find the maximizing values of μ and σ^2 , we must take the partial derivatives of $\ln(f)$ with respect to μ and σ^2 , equate them to zero, and solve the resulting two equations.
- Omitting the details, the resulting MLE's are

$$\hat{\mu} = \bar{X} \quad \hat{\sigma}^2 = \frac{\sum (X_i - \bar{X})^2}{n}$$

- The MLE of σ^2 is not the unbiased estimator, so two different principles of estimation (unbiasedness and maximum likelihood) yield two different estimators

MLEs



If θ denotes the parameter to be estimated, then estimators will be denoted by $\hat{\theta}$

- For Bernoulli: $\hat{p} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$

- For Exponential: $\hat{\lambda} = 1/\bar{X}$

- For Poisson: $\hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n}$

- For Normal: $\hat{\mu} = \frac{\sum_{i=1}^n x_i}{n}, \hat{\sigma} = \sqrt{\frac{\sum_{i=1}^n (x_i - \hat{\mu})^2}{n}}$

MLEs for the parameters of Uniform distribution



Let $x_1, x_2, \dots, x_n$ be a random sample drawn from a Uniform population with probability function

$$f(X, a, b) = 1/(b-a) \text{ for } a \leq X \leq b.$$

The likelihood function is expressed as follows:

$$L = f(x_1, a, b) f(x_2, a, b) \dots f(x_n, a, b)$$

$$= 1/(b-a) \times 1/(b-a) \times \dots \times 1/(b-a)$$

$$= 1/(b-a)^n = (b-a)^{-n} \text{ Thus } \ln L = -n \ln (b-a)$$

MLEs for the parameters of Uniform distribution

As $\ln L = -n \ln(b-a)$ and $d \ln L / da = 0$ gives $n/(b-a) = 0$ i.e. $n = 0$ (Contradiction)

Also, $d \ln L / db = 0$ gives $-n = 0$ (Contradiction)

Therefore, principle of maxima fails to give MLEs.

Now, we use order statistic principle. By ordering the sample observations, we obtain

$a \leq x_{(1)} < x_{(2)} < \dots < x_{(n)} \leq b$ since $a \leq X \leq b$

Where $x_{(1)} = \text{Min}\{x_1, x_2, \dots, x_n\}$ and $x_{(n)} = \text{Max}\{x_1, x_2, \dots, x_n\}$

As $x_{(1)}$ and $x_{(n)}$ fall nearest to a and b respectively in the inside interval $[a, b]$ then $x_{(1)}$ and $x_{(n)}$ are considered as MLEs for the respective parameters a and b using order statistic principle.

Practice Problem 1:

The number of defective hard drives produced daily by a production line can be modeled as Poisson's distribution. The counts for ten days are 7 3 1 2 4 1 2 3 1 2. Obtain max likelihood estimate of probability of 0 or 1 defectives on one day.

Solution

From the maximum likelihood estimate of λ is $\hat{\lambda} = \bar{x} = 26/10 = 2.6$. Consequently, by the invariance property, the maximum likelihood estimate of

is

$$P(X=0 \text{ or } 1) = e^{-\hat{\lambda}} + \frac{\hat{\lambda}e^{-\hat{\lambda}}}{1!}$$
$$e^{-\hat{\lambda}} + \frac{\hat{\lambda}e^{-\hat{\lambda}}}{1!} = e^{-2.6} + \frac{2.6 \cdot e^{-2.6}}{1!} = 0.267$$

There will 1 or fewer defectives on just over one-quarter of the days.

Practice Problem 2:



Consider a sample 0,1,0,0,1,0 from a binomial distribution, with the form $P[X=0]=(1-p)$, $P[X=1]=p$. Find the maximum likelihood estimate of p .

Solution :

$$L(p) = P[X=0] P[X=1] P[X=0] P[X=0] P[X=1] P[X=0]$$

$$=(1-p) p (1-p) (1-p) p (1-p)$$

$$=(1-p)^3 p^2.$$

$$\text{Log } L(p) = \log[(1-p)^3 p^2] = \log[(1-p)^3] + \log[p^2] = 3\log(1-p) + 2\log p$$

$$\frac{\partial \text{Log} L(p)}{\partial p} = 0$$

$$\text{means, } \frac{-3}{1-p} + \frac{2}{p} = 0 \Rightarrow \frac{-3p+2-2p}{p(1-p)} = 0 \Rightarrow p = 1/3$$

That is , there is 1/3 chance to observe this sample if we believe the population to be Binomial distributed .

Practice Problem 3:



Linda is using ANOVA to measure whether there is a difference between the average weekly sales of her 3 salespeople. The test will be at the 0.05 level of significance.

Weekly Sales(x) in Thousands of Dollars for 3 Treatments(T)					
Salesperson	L	7	6	7	4
	M	6	8	6	6
	N	9	8	7	10

Practice Problem 4:



Construct an analysis of variance table, and test the equality of mean weights with $\alpha = 0.05$. Each laboratory measures the tin-coating weights of 12 disks and that the results are as follows:

Laboratory A	Laboratory B	Laboratory C	Laboratory D
0.25	0.18	0.19	0.23
0.27	0.28	0.25	0.30
0.22	0.21	0.27	0.28
0.30	0.23	0.24	0.28
0.27	0.25	0.18	0.24
0.28	0.20	0.26	0.34
0.32	0.27	0.28	0.20
0.24	0.19	0.24	0.18
0.31	0.24	0.25	0.24
0.26	0.22	0.20	0.28
0.22	0.29	0.21	0.22
0.28	0.16	0.19	0.21

Practice Problem 5:



A clinical trial on Iron intake of four groups of patients (mg). Test at 5% level of significance is there any significant difference in mean iron intake among ten groups of patients as well as between three trimesters?

Trimester	Group means for iron intake									
	G ₁	G ₂	G ₃	G ₄	G ₅	G ₆	G ₇	G ₈	G ₉	G ₁₀
I	11.5	19.5	18.5	12.5	18.5	16.5	26.5	18.5	16.0	24.5
II	27.0	28.0	22.0	21.0	15.0	19.5	20.0	26.0	30.0	28.5
III	28.0	30.0	26.0	30.0	24.5	28.5	26.0	30.0	27.0	25.5

Practice Problem 6:

An experiment was performed to determine the effect of four different chemicals on the strength of a fabric. These chemicals are used as part of the permanent press finishing process. Five fabric samples were selected, and a randomized complete block design was run by testing each chemical type once in random order on each fabric sample. The data are shown in Table.

Test for differences in means using an ANOVA with $\alpha = 0.01$.

Chemical Type	Fabric Sample				
	1	2	3	4	5
1	1.3	1.6	0.5	1.2	1.1
2	2.2	2.4	0.4	2.0	1.8
3	1.8	1.7	0.6	1.5	1.3
4	3.9	4.4	2.0	4.1	3.4

Practice Problem 7:



The following data represents the number of hours of relief provided by four different brands of headache tablets administered to 20 subjects experiencing fever of 38°C or more (100.4° Fahrenheit). Does this data indicate any significant differences in means of all four groups. Is your inference remain same at 1% and 5% level of significance

Brand A	Brand B	Brand C	Brand D
5.2	9.1	3.2	2.4
4.7	7.1	5.8	3.4
8.1	8.2	8.2	4.1
6.2	6.0	3.1	5.0
3.0	9.1	7.2	4.0

Practice Problem 8:



Three different machines are being considered for use in manufacturing rubber seals. The machines are being compared with respect to tensile strength of the product. A random sample of six seals from each machine is used to determine whether the mean tensile strength varies from machine to machine. The following are the tensile strength measurements in kilograms per square centimeter $\times 10^{-1}$. Examine whether there is any difference in means of all the three machines using 5% level of significance

Machine 1	Machine 2	Machine 3
17.5	16.4	12.2
16.9	19.2	16.2
15.8	17.7	13.6
18.6	15.4	15.4
20.5	18.9	15.7
14.6	13.2	16.1

IMP Note to Self



Work Integrated Learning Programmes



Thank You

Work Integrated Learning Programmes